Name: Surabhi P
Id: COMETFWC032
Date: 28 August 2025

`https://github.com/surabhi84ard/Mini/tree/8e9042b13304f4b4923d330711d1eb424065bccb/Assembly`

Abstract

This manual shows how to use the 7474 D-Flip Flop ICs in a sequential circuit to realize a decade counter.

1 Components

Component	**Value**	**Quantity**
Breadboard	–	1
Resistor	$\geq 220\Omega$	1
Arduino	Uno	1
Seven Segment Display	Common Anode	1
Decoder	7447	1
Flip Flop	7474	2
Jumper Wires	–	20

Table 1: Required Components

2 7474 Pin Diagram

Pin	Function
14	Vcc
13	CLK2
12	D2
11	Q2
10	–
9	–
8	–
7	–
6	–
5	CLK1
4	D1
3	Q1
2	–
1	GND

3 Decade Counter Setup

3.1 Arduino to Flip-Flop and Decoder Connections

Arduino Pin	Connected To
D6	CLOCK
D7	W
D8	X
D9	Y
D2	Z
D3	A
D4	B
D5	C
D13	D

Table 2: Arduino to Logic Pins

3.2 Decoder Connections

4 Logic Flow

The 7474 flip-flops are used to generate a 4-bit binary counter. The output is incremented on each clock pulse and decoded by the 7447 to drive the seven segment display.

Decoder Pin	Connected To
7	W
1	X
2	Y
6	Z
16	Vcc

Table 3: 7447 Decoder Pin Mapping

4.1 Truth Table

CLOCK	W	X	Y	Z	A	B	C	D
0	0	0	0	0	0	0	0	1
1	0	0	0	1	0	0	1	0
2	0	0	1	0	0	1	1	0
3	0	0	1	1	1	0	0	0
4	0	1	0	0	1	0	1	0
5	0	1	0	1	1	1	0	0
6	0	1	1	0	0	0	1	1
7	0	1	1	1	0	1	0	0
8	1	0	0	0	1	0	0	0
9	1	0	0	1	0	0	0	0

Table 4: Decade Counter Output

Note